

Paper Review:

Valid Model-Free Spatial Prediction

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4 December 2024

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Discussion

- For complex processes observed across a vast and diverse spatial domain, the typical assumptions of normality and stationarity may be questionable.
- When such assumptions are not met, model-based prediction intervals are at risk of misspecification bias that can negatively affect their validity.
- While there are many spatial prediction methods for addressing nonstationarity and non-Gaussianity, they typically involve intensive computations.

Contribution of the Paper

- This paper proposes a novel approach for model-free nonparametric spatial prediction based on the **conformal prediction** machinery.
- Conformal prediction, first developed by Vovk, Gammerman, and Shafer in 2005, offers provably valid prediction intervals without specification of a statistical model.
- The crucial assumption behind conformal prediction is the exchangeability of data.
- The authors provide theoretical proofs for **local approximate exchangeability**, which holds in a wide range of settings, and introduce a **local spatial conformal prediction algorithm (LSCP)** that yields valid prediction intervals.

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Notations

- Let $Y_i \in \mathbb{R}$ and $X_i \in \mathbb{R}^d$ be the observable pairs at spatial location $s_i \in \mathcal{D} \subseteq \mathbb{R}^2$.
- Assume only a single observation is made at each location, and thus we write $Y_i = Y(s_i)$ and $X_i = X(s_i)$.
- Data points: $Z_i = (s_i, X_i, Y_i)$, $i = 1, 2, \dots$.
- Dataset: $Z^{n+1} = (Z_1, \dots, Z_{n+1})$. The goal is to predict Y_{n+1} , where Z^n will be observed and (s_{n+1}, X_{n+1}) will be given.
- Write $z_{(i)}^{n+1} = \{z_1, \dots, z_{n+1}\} \setminus \{z_i\}$.

- Geostatistical analysis often assumes that the data follow a Gaussian process model $Y_i = X_i^T \beta + \theta_i + \varepsilon_i$, where
 - ε_i are i.i.d. $N(0, \tau^2)$ errors,
 - θ is a mean-zero Gaussian process with isotropic covariance function $C(\theta_i, \theta_j) = \sigma^2 \rho(d_{ij})$, $d_{ij} = \|s_i - s_j\|$.
- Based on the observed data, we can obtain the estimates of the regression and spatial covariance parameters, $\hat{\beta}$ and $\hat{\Theta}$.
- The predictive distribution of Y_i is normal with mean $\hat{\mu}_{n+1,i}(s_i, x_i)$ and variance $\hat{\sigma}_{n+1,i}(s_i, x_i)$, and the standardized residuals are expressed as

$$e_{n+1,i} = \frac{y_i - \hat{\mu}_{n+1,i}(s_i, x_i)}{\hat{\sigma}_{n+1,i}(s_i, x_i)}, \quad i = 1, \dots, n+1. \quad (1)$$

We briefly review the original conformal prediction framework for nonspatial data. Recall that conformal prediction relies on a minimal yet crucial assumption: the *exchangeability* of data.

Definition (Exchangeability)

Let $(Z_1, \dots, Z_n, \dots)$ follow a joint distribution P . The data points $(Z_1, \dots, Z_n, \dots)$ are **exchangeable** if their joint distribution is unchanged under permutations, i.e., for any permutation ξ ,

$$(Z_1, \dots, Z_n, \dots) \stackrel{d}{=} (Z_{\xi(1)}, \dots, Z_{\xi(n)}, \dots).$$

The goal of conformal prediction is to obtain a valid prediction interval for Y_{n+1} under *any exchangeable distribution* P .

Definition (Validity of Prediction Interval)

For any $\alpha \in (0, 1)$, a prediction interval $\Gamma^\alpha(Z^n; X_{n+1})$ is considered **valid** if

$$P^{n+1}\{\Gamma^\alpha(Z^n; X_{n+1}) \ni Y_{n+1}\} \geq 1 - \alpha, \quad \forall(\alpha, n, P) \quad (2)$$

where P^{n+1} is the distribution of $(Z_1, \dots, Z_n, Z_{n+1})$ under P .

Step 1. Define a **nonconformity measure** $\Delta(B, z)$.

- A nonconformity measure $\Delta(B, z)$ is a function that takes two arguments: a "bag" B that consists of a finite collection of data points, and a single data point z .
- It measures how closely z represents the data points in B , i.e. how well z *conforms* to z .
- For example, we can define $\Delta(B, z) = d(\pi_B, z)$, where π_B is the predicted value obtained by applying the prediction rule π to B , and thus $d(\pi_B, z)$ is the discrepancy between π_B and z .

Step 2. Transform the data via Δ .

- Augment the observed data $z^n = (z_1, \dots, z_n)$ with a provisional value $z_{n+1} = (x_{n+1}, y_{n+1})$.
- Then, define

$$\delta_i = \Delta(z_{(i)}^{n+1}, z_i), \quad i = 1, \dots, n+1.$$

- Note that δ_{n+1} is special because it compares the actual observed data with this provisional value of the unobserved future observation.

Conformal Prediction Procedure

Step 3. Compute the plausibility of the provisional value z_{n+1} :

$$p(y_{n+1} \mid z^n, x_{n+1}) = \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\{\delta_i \geq \delta_{n+1}\}. \quad (3)$$

This induces a mapping $\tilde{y} \mapsto p(\tilde{y} \mid z^n, \tilde{x})$, which we will refer to as the **plausibility contour**.

Step 4. Compute the set $\Gamma^\alpha(z^n; \tilde{x})$:

$$\Gamma^\alpha(z^n; \tilde{x}) = \{\tilde{y} : p(\tilde{y} \mid z^n, \tilde{x}) > \alpha\}, \quad (4)$$

which yields a valid $100(1 - \alpha)\%$ prediction interval.

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We first consider a direct application of the original conformal algorithm, which we will refer to as the **Global Spatial Conformal Prediction (GSCP)** algorithm.

- This method assigns equal weight to nonconformity across all spatial locations in the plausibility contour evaluation, which is why it is named *global*.
- The only modification to the original conformal algorithm is allowing the nonconformity measure to account for spatial dependence.

Define the **nonconformity measure** as

$$\delta_i = \Delta(z_{(i)}^{n+1}, z_i) = \left| \frac{y_i - \hat{\mu}_{n+1,i}(s_i, x_i)}{\hat{\sigma}_{n+1,i}(s_i, x_i)} \right|, \quad i = 1, \dots, n+1 \quad (5)$$

where $\hat{\mu}_{n+1,i}(s_i, x_i)$ and $\hat{\sigma}_{n+1,i}(s_i, x_i)$ are the mean response and the standard error estimates at s_i based on the data in $z_{(i)}^{n+1}$.

- Any reasonable choice of $(\hat{\mu}, \hat{\sigma})$ can be used, including deep learning predictions and Kriging predictions.
- We use the Kriging estimates, so δ_i is a standardized Kriging residual $|e_i|$. This is due to the computational advantages of Kriging.

Then, we define a **plausibility contour** in a similar manner. For provisional values $(s_{n+1}, x_{n+1}, y_{n+1})$, we have:

$$p(y_{n+1}|z^n; s_{n+1}, x_{n+1}) = \frac{1}{n+1} \sum_{i=1}^n \mathbf{1}\{\delta_i \geq \delta_{n+1}\}, \quad (6)$$

and the corresponding $100(1 - \alpha)\%$ prediction interval for Y_{n+1} , $\Gamma^\alpha(z^n; s_{n+1}, x_{n+1})$ is just an upper level set of the plausibility contour.

- Note that the validity of $\Gamma^\alpha(z^n; s_{n+1}, x_{n+1})$ does not depend on model specification, nor on our choice of the δ_i 's.
- *However, we do require the exchangeability of the data.* When can this assumption hold in the presence of spatial dependence?

Exchangeability holds when the **spatial locations are sampled independently and identically distributed (iid)**.

Lemma 1.

If the spatial locations $S_1, S_2, \dots$ are iid, then $Z_1, Z_2, \dots$, with $Z_i = (S_i, X(S_i), Y(S_i))$ is an exchangeable sequence.

Under this assumption, the validity results follow immediately.

Theoretical Validity of GSCP

Theorem 1.

Let (X, Y) be a stochastic process over $\mathcal{D}$ and let $S_1, S_2, \dots$ be iid draws in $\mathcal{D}$. Define the coverage probability function as

$$c(\alpha, n, P) = P^{n+1}(\{Y_i \in \Gamma^\alpha(Z^n; X_{n+1}, S_{n+1})\}).$$

Then, the proposed GSCP is **valid** in the sense that

$$c(\alpha, n, P) \geq 1 - \alpha, \quad \forall(\alpha, n, P), \quad (7)$$

where P^{n+1} is the joint distribution of Z^{n+1} under P . Moreover, if $\delta_1, \dots, \delta_{n+1}$ have a continuous distribution, then the following holds:

$$c(\alpha, n, P) \leq 1 - \alpha + (n+1)^{-1}, \quad \forall(\alpha, n, P). \quad (8)$$

This provides a marginal validity result, but does not ensure conditional validity when $S_{n+1} = s^*$ with s^* being a fixed spatial location.

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Local Spatial Conformal Prediction

For a valid prediction **at a fixed location** s^* , we propose a **local spatial conformal prediction (LSCP)** approach that is based on only those data points in the neighborhood of s^* .

- Fix an integer $m > 0$ and select a neighborhood around s^* that contains m many locations s_{ij} , $j = 1, \dots, m$.
 - Without structural assumptions about the response process, such as stationarity, data points far from s^* are not obviously relevant to prediction at s^* .
 - When the infill asymptotic is appropriate, m could be taken to be large.
- With an abuse of notation, write $Z_i = (s_i, X_i, Y_i)$, for $i = 1, \dots, m$, for the points in the neighborhood of s^* , and set $s_{m+1} = s^*$.

We proceed exactly as we did before.

- **Nonconformity measure:**

$$\delta_i = \Delta(z_{(i)}^{m+1}, z_i), \quad i = 1, \dots, m+1$$

- **Plausibility contour:**

$$p(y_{m+1}|z^m, s^*, x_{m+1}) = \frac{1}{m+1} \sum_{i=1}^{m+1} \mathbf{1}\{\delta_i \geq \delta_{m+1}\} \quad (9)$$

This yields a $100(1 - \alpha)\%$ prediction interval $\Gamma_{s^*}^\alpha(Z^m; X_{m+1})$ for $Y_{m+1} = Y(s^*)$, depending on Z^m and the observed $X_{m+1} = X(s^*)$.

Theoretical Validity of LSCP

The theoretical validity of LSCP hinges on a definition of **local exchangeability**, which allows for an *asymptotically* valid prediction interval. We first introduce a set of mild assumptions.

- Let $\mathcal{D}$ be a compact spatial domain, and define the localized version of the process T , relative to a location $s^* \in \mathcal{D}$ and a radius r :

$$\tilde{T}_r(u) = T(s^* + ru), \quad u \in \mathcal{U} = \{u \in \mathbb{R}^2 : \|u\| \leq 1\} \quad (10)$$

- Assume that T can be decomposed as

$$T(s) = \Psi(L(s), E(s)), \quad s \in \mathcal{D}, \quad (11)$$

where L, E are independent real-valued stochastic processes, L is spatial, and E is nonspatial. Ψ is a deterministic continuous function.

Theoretical Validity of LSCP

Further assume that the processes L, E in the decomposition (11) satisfy the following conditions:

- L is L_2 -continuous at s^* , in the sense that $\mathbb{E}||\tilde{L}_r(u) - \tilde{L}_0(u)||^2 \rightarrow 0$ as $r \rightarrow 0$ for any $u \in \mathcal{U}$;
- E is locally iid at s^* , in the sense that $\tilde{E}_r$ converges in distribution to an iid process $\tilde{E}_0$ as $r \rightarrow 0$.

While this formulation is too abstract to be immediately useful, it is satisfied by many well-known spatial models.

- An example is the generalized spatial linear model,

$$\psi_Y(\ell, e) = H_{g(\ell)}^{-1}(\Phi^{-1}(e)).$$

Now, we are ready to establish the **local exchangeability** result:

Proposition 1.

Suppose that T can be decomposed as in (11) where L is L_2 -continuous at s^* , E is locally iid at s^* , and L, E are independent.

Then the localized process $\tilde{T}_r$ in (10) converges in distribution as $r \rightarrow 0$, and the limit is an exchangeable process in the sense that its finite-dimensional distributions are exchangeable.

- Under an *infill asymptotic regime*, we can use the results from Proposition 1.
 - Such a regime is natural (and necessary) in cases without structural assumptions, where it is impossible to learn the local features if data are not concentrated.
- For a fixed m , the radius r of neighborhood containing m points tends to $r_n = (m/n)^{1/2} \rightarrow 0$ as $n \rightarrow \infty$.
- Hence, it follow from Proposition 1 that the joint distribution of Y at m -many locations around s^* would be **approximately exchangeable**.

Theoretical Validity of LSCP

Therefore, we obtain an **approximately valid** conformal prediction interval:

Theorem 2.

Consider an infill asymptotic regime with n spatial locations in the bounded domain $\mathcal{D} \subseteq \mathbb{R}^d$, with $n \rightarrow \infty$. Fix an integer $m > 0$ and let $r = r_n \rightarrow 0$ such that the m closest data points to s^* are in the neighborhood $B(s^*, r)$. Under the assumptions of Proposition 1, the nonconformity measure Δ is a continuous function of its inputs, and if the limiting distribution is continuous, then the LSCP prediction intervals are asymptotically valid at s^* in the sense that

$$\lim_{n \rightarrow \infty} P^{m+1} \{Y_{m+1} \in \Gamma^\alpha(Z^m; s^*, x_{m+1})\} = 1 - \alpha + O(m^{-1}). \quad (12)$$

The Smoothed LSCP Algorithm (sLSCP)

In practice, completely disregarding the contribution of the observations outside the m nearest neighbors may be unsatisfactory. The **Smoothed LSCP Algorithm (sLSCP)** is a modification of LSCP that assigns weights based on their distance to the target location s^* :

$$w_i \propto f(d_i) = \frac{\exp(-d_i^2/2\eta^2)}{1 + \sum_{j=1}^n \exp(-d_j^2/2\eta^2)}, \quad i = 1, \dots, n+1$$

where $d_i = \|s_i - s^*\|$ and $d_{n+1} = 0$.

With such weights, the plausibility contour is given by

$$p(y_{n+1} | Z_{(n+1)}^{n+1}, s^{n+1}, x_{n+1}) = \sum_{i=1}^{n+1} w_i \mathbf{1}\{\delta_i \geq \delta_{n+1}\}. \quad (13)$$

The Smoothed LSCP Algorithm (sLSCP)

- Due to computational issues, it is recommended to use only the $M \ll n$ neighboring data points in the computation of the plausibility contour.
- As a rule of thumb, M could be selected to roughly include all observations within $2\eta + r^*$ radius of s^* , where r^* is selected to include a sufficient number of neighbors.
- Other kernels can also be used. Kernel hyperparameters (such as η) are selected via cross-validation.

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Through a simulation study, we aim to observe:

1. Whether conformal algorithms can perform as well as standard methods when assumptions are met,
2. Their predictive ability and efficiency in non-stationary, non-Gaussian settings.

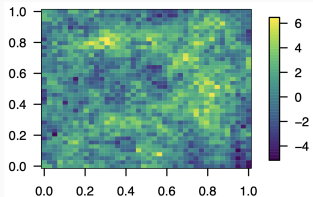
1. Data generation

We generate 8 different scenarios, using a latent Gaussian process $Z(s)$ and a white noise process $E(s)$.

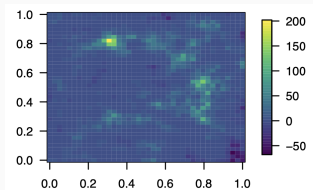
- $Z(s)$: Mean-zero stationary Gaussian process with Matérn covariance function, with $\sigma^2 = 3$, $\phi = 1$, $\kappa = 0.7$.
- $E(s)$: White noise generated with standard normal distribution.
- Data are sampled on the $N \times N$ equally-spaced grid of points in the unit square, with $N = 20$ or $N = 40$.

Scenario	Description
Scenario 1	Stationary & Gaussian.
Scenarios 2-4	Stationary, but non-Gaussian.
Scenarios 5-8	Nonstationary in terms of variances or mean.

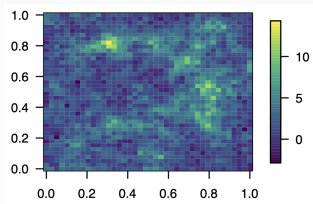
Simulation



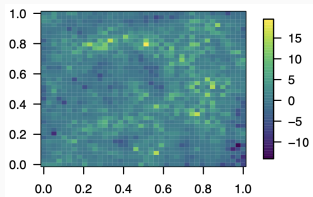
Scenario 1. $Y(s) = Z(s) + E(s)$



Scenario 2. $Y(s) = Z(s)^3 + E(s)$

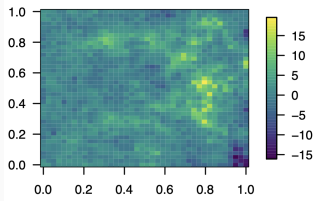


Scenario 3. $Y(s) = q \left[\Phi \left(\frac{Z(s)}{\sqrt{3}} \right) \right] + E(s)$



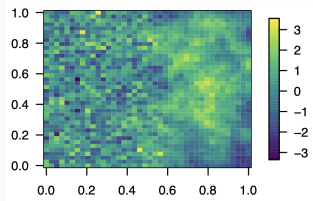
Scenario 4. $Y(s) = \sqrt{3}Z(s)|E(s)|$

Simulation



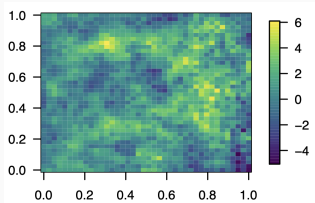
Scenario 5.

$$Y(s) = \text{sign}\{Z(s)\}|Z(s)|^{s_x+1} + E(s)$$

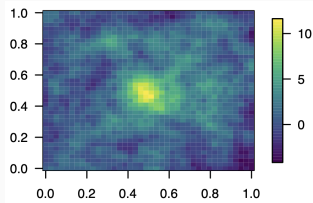


Scenario 6.

$$Y(s) = \sqrt{\omega(s)/3}Z(s) + \sqrt{1 - \omega(s)}E(s)$$



Scenario 7. $Y(s) = Z(s) + s_x E(s)$



Scenario 8. $Y(s) = Z(s) + 10e^{-50\|s-c\|^2}$

2. Prediction methods and metrics

We compare four methods at a prediction level of $\alpha = 0.1$:

1. **GSCP**
2. **LSCP**: Gaussian kernel, $\eta = 0.1$
3. **Kriging**: Empirical variogram methods
4. **Local Kriging (laGP)**: Proposed by Gramacy and Apley in 2015, a method that dynamically defines the support of a Gaussian predictor

Each scenario is simulated 100 times with a 90:10 train-test split.

We use the (1) average coverage, (2) average interval width, and (3) average interval score to evaluate the performance.

3. Results

- Scenario 1: $\text{GSCP} \approx \text{LSCP} \approx \text{Kriging}$.
 - Conformal methods are competitive, even when the data-generating mechanism aligns with that of Kriging.
- Scenarios 2 - 4: $\text{GSCP} \approx \text{LSCP} \approx \text{Kriging} > \text{laGP}$.
- Scenarios 5 - 6: LSCP performs best in these nonstationary scenarios, where spatial variance varies by location.
- GSCP and LSCP are comparable in performance, even in stationary settings.
- LSCP outperforms laGP in nonstationary settings by providing adaptive prediction interval width and valid coverage.

Simulation Results

		$N = 20$			$N = 40$		
Scen	Method	Cov90	Width	IntScore	Cov90	Width	IntScore
1	GSCP	0.906	4.67	5.78	0.897	4.00	5.06
	LSCP	0.890	4.57	5.95	0.891	3.99	5.12
	Kriging	0.912	4.73	5.78	0.888	3.90	5.07
	LaGP	0.877	4.78	6.50	0.879	4.27	5.63
2	GSCP	0.895	33.62	67.16	0.897	22.12	43.80
	LSCP	0.896	31.05	58.37	0.910	21.74	36.47
	Kriging	0.931	44.07	69.38	0.924	27.75	44.80
	LaGP	0.913	40.57	69.38	0.928	31.28	47.24
3	GSCP	0.908	4.79	6.32	0.895	4.05	5.27
	LSCP	0.893	4.65	6.27	0.893	4.03	5.24
	Kriging	0.919	4.95	6.33	0.887	3.96	5.28
	LaGP	0.883	4.92	6.83	0.880	4.29	5.77
4	GSCP	0.902	7.06	11.06	0.895	6.25	10.25
	LSCP	0.892	6.84	11.23	0.895	6.08	9.74
	Kriging	0.918	7.70	11.12	0.908	6.71	10.26
	LaGP	0.898	7.40	11.83	0.901	6.68	10.51

Table 1: Average performance comparison for Scenarios 1 - 4.

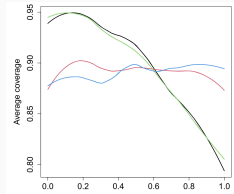
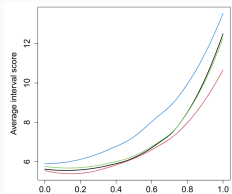
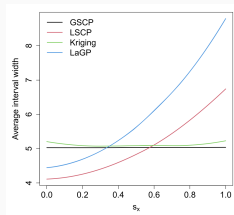
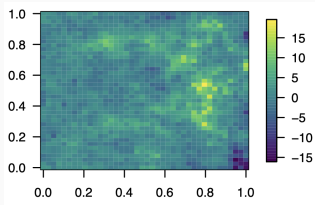
Simulation Results

		$N = 20$			$N = 40$		
Scen	Method	Cov90	Width	IntScore	Cov90	Width	IntScore
5	GSCP	0.900	6.51	9.40	0.898	5.03	7.11
	LSCP	0.887	6.36	9.06	0.892	5.05	6.75
	Kriging	0.924	7.18	9.52	0.897	5.11	7.15
	LaGP	0.887	7.20	10.42	0.891	5.98	8.08
6	GSCP	0.894	2.78	3.69	0.896	2.63	3.60
	LSCP	0.878	2.66	3.47	0.895	2.38	3.06
	Kriging	0.897	2.78	3.69	0.888	2.54	3.61
	LaGP	0.865	2.58	3.60	0.869	2.32	3.21
7	GSCP	0.905	3.63	4.55	0.896	2.77	3.71
	LSCP	0.888	3.53	4.59	0.896	2.70	3.46
	Kriging	0.915	3.77	4.55	0.889	2.70	3.72
	LaGP	0.869	3.92	5.31	0.881	3.17	4.14
8	GSCP	0.906	3.04	3.74	0.899	1.93	2.45
	LSCP	0.880	3.00	3.91	0.895	1.92	2.46
	Kriging	0.928	3.25	3.78	0.915	2.05	2.48
	LaGP	0.863	3.39	4.64	0.871	2.41	3.20

Table 2: Average performance comparison for Scenarios 5 - 8.

Simulation Results

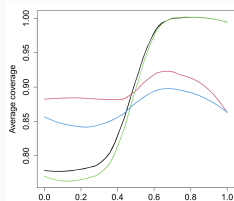
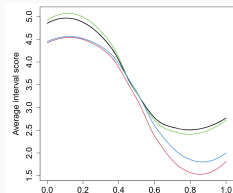
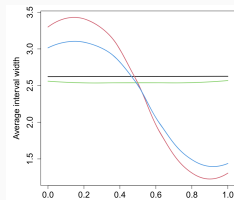
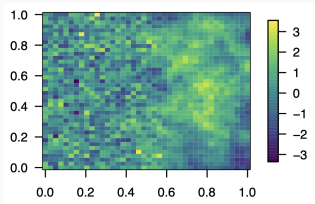
Scenario 5. $Y(s) = \text{sign}\{Z(s)\}|Z(s)|^{s_x+1} + E(s)$



The process is non-Gaussian to the east; LSCP and laGP capture the changes.

Simulation Results

Scenario 6. $Y(s) = \sqrt{\omega(s)/3}Z(s) + \sqrt{1 - \omega(s)}E(s)$, $\omega(s) = \Phi(\frac{s_x - 0.5}{0.1})$



The correlation is stronger in the east. LSCP performs best among all methods.

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We demonstrate the performance of conformal prediction using the **canopy height data** in Bonanza Creek Experimental Forest, AK.

- Total $n = 1,723,137$ observations.
- Data exhibit clear nonstationarity and non-Gaussianity, including small heterogeneous areas.

	Width	Cov90	IntScore
LSCP	2.87	87.9%	4.33
Kriging	5.44	96.6%	6.81
laGP	5.04	91.1%	6.63

Table 3: Performance comparison for canopy height data.
Metrics averaged over 10,000 test locations.

Real Data Analysis

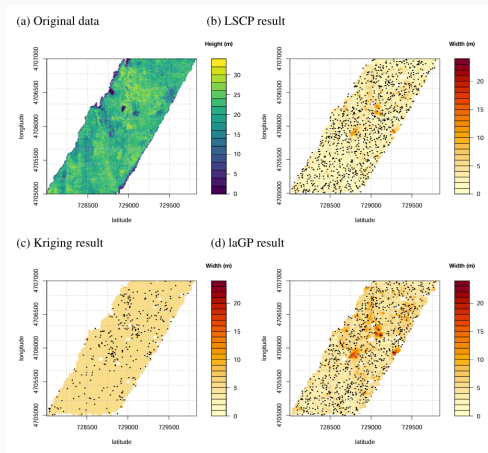


Figure 1: (a) Heatmap of the canopy height data, (b)-(d) Prediction interval width (color) and locations not covered (points).

Observations:

- While the empirical coverage for all three methods is close to the nominal level, the LSCP algorithm stands out in terms of efficiency.
- Both LSCP and laGP produce locally adaptive intervals, whereas Kriging exhibits homogeneity. This results in clustered areas where observations fall outside the prediction intervals.
- LSCP generally provides narrower intervals than laGP.

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Key outcomes:

- This work is among the first to apply the classical conformal algorithm to nonexchangeable data.
- The proposed algorithms are model-free, in the sense that their theoretical validity does not depend on model specification.
- The algorithm retains the computational advantages of model-based methods like Kriging, leveraging them solely for defining nonconformity measures.
- The R package `scp` is also available.

Future research directions:

- Extending the method to handle spatial processes with discrete observations, such as binary or count data.
- Developing simultaneous prediction techniques.
- Expanding the approach to spatio-temporal data, including time series analysis.